

The relation of passage of gas an abdominal bloating to colonic gas production.

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OBJECTIVE: To determine the relation of gas passage and abdominal bloating to the production of gas in the colon. DESIGN: Randomized, double-blind, crossover study of gaseous symptoms during a 1-week period. SETTING: A Veterans Affairs medical center. PARTICIPANTS: 25 healthy medical center employees. INTERVENTION: Participants' diets were supplemented with either a placebo (10 g of lactulose, a nonabsorbable sugar), psyllium (a fermentable fiber), or methylcellulose (a nonfermentable fiber). MEASUREMENTS: All participants were polled for gaseous symptoms (including number of gas passages, impression of increased rectal gas, and abdominal bloating), and five were examined for breath hydrogen excretion. RESULTS: Participants passed gas 10 +/- 5.0 times per day (mean +/- SD) during the placebo period. A significant increase in gas passages (to 19 +/- 12 times per day) and a subjective impression of increased rectal gas were reported with lactulose but not with either of the two fiber preparations. Breath hydrogen excretion, an indicator of hydrogen production in the colon, did not increase after ingestion of either of the fibers. However, a statistically significant ($P < 0.05$) increase in feelings of abdominal bloating (which the participants perceived as excessive gas in the bowel) was reported with both fiber preparations and with lactulose. CONCLUSIONS: The physician should distinguish between excessive gas (which indicates excessive gas production) and feelings of bloating (which are usually unrelated to excessive gas production). Treatment of the former consists of limiting the supply of fermentable material to the colonic bacteria. Symptoms of bloating usually indicate the irritable bowel syndrome, and therapy should be directed accordingly.